SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1001	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	T. Tarun	**Reg. No.:**	1924260001
Date:		**Duration:**	60 minutes

Code of Conduct

I,T. Tarun(1924260001) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T. Tarun

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

REAL-TIME WATER MANAGEMENT SYSTEM

1. Identify the Project Vision and Stakeholders

Define the project's vision, objectives, and identify all key stakeholders involved.

1.1 Project Vision

To build a reliable, accessible real-time water management platform that transforms sensor readings into actionable information. The system will help facility teams detect low water levels, potential overflow, unusual consumption, and possible leakage early, while giving authorized users transparent access to water-use information.

1.2 Project Objectives

The main objectives of the Real-Time Water Management System are:

- To develop an intelligent water management system that monitors water levels, flow rates, and consumption in real time.
- To minimize water wastage by detecting leakage, overflow, and abnormal water usage.

- To integrate IoT sensors for continuous monitoring and accurate collection of water-related data.
- To provide automatic alerts and notifications when water levels reach critical limits or leakage is detected.
- To develop a centralized dashboard for monitoring water availability, consumption, and system performance.
- To maintain a database for storing water usage records, sensor readings, and historical information.
- To improve water distribution efficiency by analyzing consumption patterns and identifying areas with high demand.
- To reduce manual monitoring and improve operational efficiency through automated data collection.
- To support data-driven decision-making through real-time analytics and water consumption reports.
- To promote sustainable water resource management by improving conservation, reliability, and efficient utilization of water.

1.3 Key Stakeholders and Responsibilities

Stakeholder	Role / responsibility
Water management authority	Reviews water availability, reports and resource planning.
Facility manager	Monitors tanks and oversees daily operations.
Maintenance personnel	Investigates alerts, records findings and closes incidents.
Residents / consumers	Views permitted consumption information and notices.
System administrator	Manages accounts, roles, devices and system configuration.
IoT / sensor operator	Installs, calibrates and checks sensor connectivity.
Development and QA team	Builds, tests and maintains the software and integrations.

1.4 Scope and Assumptions

The first release covers a web dashboard, simulated or connected sensor readings, alerts, and incident tracking. Automated valve control and citywide billing are outside the first sprint. Sensor readings are assumed to include a device identifier, value, unit and timestamp.

2. Create Epics and User Stories

Break down the requirements into Epics and formulate well-structured User Stories using the format: As a <user>, I want <feature>, so that <benefit>.

2.1 Epic: Live Water Monitoring

US-01: As a facility manager, I want to view the latest tank water level with a timestamp, so that I can identify shortage or overflow risks.

US-02: As a facility manager, I want to view incoming flow measurements, so that I can detect unexpected changes in supply.

2.2 Epic: Alerts and Incident Response

US-03: As a maintenance technician, I want to receive an alert when readings cross configured limits, so that I can investigate promptly.

US-04: As a maintenance technician, I want to acknowledge and close alerts with notes, so that actions are traceable.

2.3 Epic: Usage and Reporting

US-05: As a resident, I want to see my permitted daily consumption summary, so that I can understand usage trends.

US-06: As a facility manager, I want to export a weekly water-use report, so that I can compare periods and plan conservation.

2.4 Epic: Administration

US-07: As a system administrator, I want to register sensors and assign them to tanks, so that incoming readings are associated with the correct location.

US-08: As a system administrator, I want to control user permissions, so that sensitive operational data is visible only to authorized people.

3. Define Acceptance Criteria

Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

3.1 US-01 – Live Tank Level

Given a registered sensor sends a valid reading, when the backend receives it, then the dashboard shows the latest percentage and timestamp within 10 seconds. Given no new reading for 5 minutes, when the dashboard refreshes, then it marks that sensor as “Offline / stale” rather than showing the old value as current.

3.2 US-03 – Threshold Alerts

Given a tank has a low-level threshold of 20%, when a valid reading falls below 20%, then the system creates one active low-level alert for that tank and displays it in the alert panel. Given the condition remains active, when more readings arrive, then duplicate active alerts are not created.

3.3 US-04 – Incident Acknowledgement

Given an open alert is visible to a maintenance user, when the user selects “Acknowledge”, then the system stores the user, time and updated status. Given an acknowledged alert, when the user adds a resolution note and closes it, then the incident history records the note and closing time.

3.4 US-07 – Sensor Registration

Given an administrator is signed in, when they submit a unique sensor ID and valid tank assignment, then the device appears in the sensor list. Given an existing sensor ID, when it is submitted again, then the system rejects the duplicate with a clear message.

4. Prioritize the Product Backlog

Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.

4.1 Prioritized Product Backlog

Rank	Story	Priority	Reason/ dependency
1	US-07 Sensor registration	Must	Needed to identify readings by tank.
2	US-01 Live tank level	Must	Core visibility; depends on registered sensors.
3	US-03 Threshold alerts	Must	Early warning; depends on readings.
4	US-04 Incident tracking	Must	Ensures alerts lead to accountable action.
5	US-08 User permissions	Should	Controls access before broader rollout.
6	US-02 Flow monitoring	Should	Improves diagnosis after level monitoring.
7	US-05 Consumption summary	Could	Useful once sufficient readings exist.
8	US-06 Weekly export	Could	Depends on historical usage records.

4.2 Prioritization Method

MoSCoW prioritization is used: Must-have features establish safe monitoring and response; Should-have features strengthen operations; Could-have features add reporting value. Technical dependencies are considered before assigning sprint work.

5. Plan the Sprint Backlog

Select high-priority user stories for the first sprint and divide them into actionable development tasks.

5.1 Sprint 1 Overview

Sprint duration: 2 weeks (10 working days). Team assumption: four student contributors. Sprint 1 includes US-07, US-01, US-03 and US-04 using sample sensor messages where physical devices are unavailable.

5.2 Sprint Backlog and Development Tasks

Story	Actionable tasks	Expected result
US-07	Design device/tank data model; build registration form; validate unique IDs; test save and retrieval.	Administrator can register and view sensors.
US-01	Create reading API; store timestamps; design level dashboard; show stale-data indicator; test updates.	Current tank level visible with time.
US-03	Store configurable threshold; evaluate incoming readings; create alert record; prevent duplicates; test boundaries.	Low-level warning appears reliably.
US-04	Build alert list; add acknowledge/close actions; store audit trail; test role and status changes.	Maintenance can track an alert to closure.

5.3 Sprint Workflow and Definition of Done

Sprint Workflow:

The Real-Time Water Management System follows an Agile sprint workflow to ensure systematic development, continuous testing, and timely completion of project tasks.

- Sprint Planning: Identify high-priority user stories and assign development tasks to team members.
- Requirement Analysis: Analyze the requirements for water level monitoring, leakage detection, and alert notifications.
- System Design: Design the user interface, database structure, and IoT sensor integration.

- Development: Implement water monitoring features, sensor data processing, and dashboard functionalities.
- Testing: Verify sensor readings, water level updates, leakage detection, and notification accuracy.
- Sprint Review: Demonstrate completed features and collect feedback from stakeholders.
- Sprint Retrospective: Evaluate team performance, identify challenges, and suggest improvements for the next sprint.

Definition of Done (DoD):

A user story is considered completed when the following conditions are satisfied:

- All specified functional requirements are implemented successfully.
- The developed features satisfy the defined acceptance criteria.
- Water level and flow sensor readings are displayed correctly.
- Leakage detection and overflow alerts function as expected.
- The dashboard displays updated water monitoring information.
- Sensor data is stored securely in the database.
- The system is tested to identify and resolve errors.
- The completed functionality is reviewed and approved by the project team.
- All necessary technical documentation is updated.
- The completed feature is ready for demonstration and integration with other system modules.

6. Estimate Effort Using Story Points

Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

6.1 Estimation Technique

Story points are used in Agile software development to estimate the relative effort required to complete user stories. The estimation considers task complexity, development effort, technical challenges, and potential risks.

In the Real-Time Water Management System, story points help the development team plan tasks efficiently, distribute workloads, and complete high-priority features within the sprint.

6.2 Story Point Estimates

Story	Points	Justification
US-07	3	Simple form, validation, database mapping and tests.
US-01	5	Requires reading ingestion, dashboard updates and stale-data handling.
US-03	5	Threshold logic, deduplication and edge-case testing.

US-04	3	Status transitions, notes and audit timestamps.
Total	16	Proposed Sprint 1 workload; to be revised after team estimation.

6.3 Estimation Risks

User Story	Development Task	Story Points
US01	User login and authentication	3
US02	Real-time water level monitoring	5
US03	Water flow monitoring	5
US04	Leakage detection system	8
US05	Automatic overflow alerts	5
US06	Water monitoring dashboard	5
US07	Sensor data storage	3
US08	Water consumption reports	5
Total		39

6.4 Justification of Story Point Estimation

- User Authentication (3 Points): Requires login form development, user validation, and basic security testing.
- Water Level Monitoring (5 Points): Requires sensor integration, real-time data collection, and display functionality.
- Water Flow Monitoring (5 Points): Involves processing sensor readings and calculating water flow rates.
- Leakage Detection (8 Points): Requires abnormal flow identification, detection logic, and testing under different conditions.
- Overflow Alerts (5 Points): Requires threshold monitoring and automatic notification generation.
- Monitoring Dashboard (5 Points): Involves displaying real-time water levels, consumption details, and system status.
- Sensor Data Storage (3 Points): Requires database connectivity and reliable storage of sensor readings.
- Consumption Reports (5 Points): Requires retrieving historical data and generating water usage summaries.

7. Present the Sprint Goal and Expected Deliverables

Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

7.1 Sprint Goal

The main goal of the first sprint is to develop the essential functionalities of the Real-Time Water Management System, including water level monitoring, sensor data collection, database integration, and a basic monitoring dashboard.

The sprint aims to establish a reliable foundation for continuous water monitoring, efficient resource management, and automated system operations.

7.2 Expected Deliverables

Deliverables include: a functional web dashboard; sensor registration screen; timestamped reading storage; low-water alert rules; an incident list with acknowledgement and closure; test cases and results for selected user stories; and a short demonstration with sample data.

7.3 Sprint Review and Success Measures

The sprint is successful when all four selected stories meet their acceptance criteria during demonstration, the team can trace each alert to its source reading, and no unresolved critical defects block the core monitoring flow. Feedback from the sprint review will shape the next backlog refinement.

7.4 Proposed Next Sprint

Sprint 2 can add flow monitoring, role-based permissions, consumption summaries and reporting, depending on team velocity and stakeholder feedback.

Conclusion

The Agile plan converts the Real-Time Water Management System concept into measurable, prioritized work. By focusing the first sprint on sensor registration, live monitoring and accountable alert handling, the project delivers a practical foundation for reducing water wastage and improving maintenance response. Later sprints can extend the system using feedback and verified operational needs.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25